

Unit 22: Newton's Method

1. The idea, in plain words

Sometimes an equation like $x^4 - 2 = 0$ has a real solution, but there's no clean algebra trick to solve for it exactly (try it — you can't just factor your way to the answer). Newton's Method is a clever way to zero in on that solution numerically, getting closer and closer with each step, using nothing but tangent lines.

Picture it like this: you make a rough first guess, x_0 , for where the root (the place where the graph crosses the x-axis) might be. You're probably not exactly right — but you don't need to be. Draw the tangent line to the curve at your guess. Since the tangent line is straight, it's easy to find exactly where it crosses the x-axis — and that crossing point becomes your new, improved guess, x_1 . Repeat the process at x_1 to get an even better guess x_2 , and so on. Each round, your guess typically gets closer to the true root, often startlingly fast.

Where the formula comes from: the tangent line at x_n is $y - f(x_n) = f'(x_n)(x - x_n)$. To find where this tangent line crosses the x-axis, set $y = 0$ and solve for x — that x becomes your next guess, x_{n+1} . Working through that algebra gives you the Newton's Method formula directly.

2. Toolbox

Newton's Method formula:

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

The procedure:

1. Choose a starting guess x_0 , ideally reasonably close to the actual root, and somewhere the derivative isn't zero.
2. Compute $f(x_0)$ and $f'(x_0)$.
3. Plug into the formula to get x_1 .
4. To go further, repeat the same process using x_1 in place of x_0 to get x_2 , and so on — each new guess uses the derivative evaluated at the most recent guess, not the original one.

A quick sanity check on the formula: if $f(x_n)$ is positive and $f'(x_n)$ is positive, the correction term is subtracted, moving your guess to the left — matching the picture of a tangent line sloping up crossing the axis to the left of where you are.

3. Common mistakes

- Reusing $f'(x_0)$ for every iteration, instead of recomputing the derivative at the current guess each time. Each new step needs f' evaluated at the most recent x -value, not the original one.
- Sign errors in the subtraction. The formula is x_n minus $\frac{f(x_n)}{f'(x_n)}$ — it's easy to accidentally add instead.
- Stopping too early. If a problem asks for x_2 , you need to complete two full iterations — computing x_1 alone isn't enough.
- Mixing up $f(x)$ and $f'(x)$ in the formula, or dividing by the wrong one.
- Rounding too aggressively partway through. Carry several decimal places through your intermediate calculations, and only round your very final answer — early rounding can noticeably throw off later iterations.

4. Problem Set

For each problem, use Newton's Method to compute the requested iteration(s), starting from the given x_0 .

Warm-up

1. $f(x) = x^2 - 2$, $x_0 = 1$. Find x_1 .
2. $f(x) = x^2 - 5$, $x_0 = 2$. Find x_1 .
3. $f(x) = x^3 - 2$, $x_0 = 1$. Find x_1 .
4. $f(x) = x^2 - 10$, $x_0 = 3$. Find x_1 .
5. $f(x) = x^3 - 8$, $x_0 = 3$. Find x_1 .
6. $f(x) = x^2 - 7$, $x_0 = 3$. Find x_1 .

Standard

7. $f(x) = x^2 - 3$, $x_0 = 2$. Find x_1 and x_2 .
8. $f(x) = x^3 - 5$, $x_0 = 2$. Find x_1 and x_2 .
9. $f(x) = x^2 + x - 6$, $x_0 = 1$. Find x_1 and x_2 .
10. $f(x) = \cos x - x$, $x_0 = 1$ (radians). Find x_1 .
11. $f(x) = x^3 - x - 1$, $x_0 = 1$. Find x_1 and x_2 .
12. $f(x) = x^2 - 6$, $x_0 = 2$. Find x_1 and x_2 .

Challenge

13. $f(x) = x^4 - 2$, $x_0 = 1$. Find x_1 .
14. $f(x) = x^4 - 2$, $x_0 = 1$. Find x_1 and x_2 .
15. $f(x) = x^5 - 3$, $x_0 = 1$. Find x_1 and x_2 .
16. Explain what goes wrong with Newton's Method if you happen to choose a starting guess x_0 where $f'(x_0) = 0$. Use the formula itself to justify your answer.
17. $f(x) = \sin x$, $x_0 = 3$ (radians). Find x_1 and x_2 , and compare x_2 to the actual value of π .

Applied

18. Use Newton's Method to approximate $\sqrt{7}$ (i.e., find the positive root of $f(x) = x^2 - 7$), starting from $x_0 = 2.5$. Find x_1 .
19. A company's break-even condition (cost minus revenue) is modeled by $f(x) = x^2 - 50x + 200 = 0$. Use Newton's Method starting from $x_0 = 5$ to find x_1 , an approximation of the lower break-even production level.
20. Find the root of $f(x) = x^3 - 2x - 5$, starting from $x_0 = 2$. Find x_1 and x_2 .
21. Use Newton's Method to approximate $\sqrt[3]{10}$ (i.e., find the positive root of $f(x) = x^3 - 10$), starting from $x_0 = 2$. Find x_1 .